

Johan Varghese K

B.Tech Computer Science & Engineering, College of Engineering Trivandrum
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Profile

Computer Science undergraduate with strong academic performance and interests in software engineering, scalable systems, and applied problem solving. Experienced in working on technical projects across software development, low-level computing, and data structures, along with research exposure through collaborative academic work.

Education

College of Engineering Trivandrum B.Tech in Computer Science & Engineering CGPA: 9.76	2024 – Present
Hari Sri Vidya Nidhi School ISC (Class XII) - 96% ICSE (Class X) - 98%	2012 – 2024

Skills

- **Programming:** C, Python, Java, JavaScript
- **Tools:** Git, GitHub, Linux (WSL), VS Code
- **Web Development** React, HTML/CSS
- **Databases:** PostgreSQL, MongoDB, Cassandra
- **Core Areas:** Data Structures and Algorithms, Object-Oriented Programming, Computer Organization, Theory of Computation

Projects

Nosh n Go — Canteen Management Web Application	Group Project
• Developed a web-based canteen management system for food pre-booking, queue optimisation, and crowd handling • Contributed to frontend development and user interface implementation	
Probabilistic Data Structures Lab	Individual Project
• Implementing and analysing probabilistic data structures including Bloom Filters and Count-Min Sketch • Exploring trade-offs between memory efficiency, probabilistic accuracy, and computational performance	
Hotel Management System (Java GUI Application)	Group Project
• Built a desktop application for managing hotel bookings and restaurant operations • Contributed to GUI development and payment logic implementation	
Design and Implementation of an 8-bit Computer	Group Project
• Designed and implemented a functional 8-bit computer using discrete logic circuits • Implemented instruction execution and arithmetic operations based on Von Neumann architecture principles.	

Experience

Research Internship

IIT Kharagpur | June 2026

- Developed automation scripts and experimental workflows for benchmarking LLM inference on heterogeneous edge devices within an embedded AI security research project.
- Conducted systematic evaluation of performance and security trade-offs to support ongoing research in trustworthy edge AI systems.

IDEA Research Internship

CET | Sep – Oct 2025

- Worked on a research-oriented project under faculty guidance in the Department of Computer Science and Engineering
- Gained exposure to structured problem solving, literature review, experimentation, and technical documentation

Publications

GA-TMC: A Meta-heuristic Approach for Co-scheduling Tasks and Messages on Shared Bus-Based Heterogeneous Platforms

Journal of Systems Architecture (Elsevier) July 2026

- Co-authored and contributed to research on task and message co-scheduling in heterogeneous embedded systems
- Worked on optimisation approaches, experimentation, and technical documentation as part of the research workflow

Leadership & Activities

- Presented a civic-tech solution at Allianz Risk Consulting Compass-360 Expo, engaged with industry professionals
- IPR Lead, IEDC CET — Innovation and Entrepreneurship Development Centre
- Administrative Committee Member, FOSS CET — campus free and open-source software community
- Member, Web Development Team, Glitch CET — game development community
- Member, Research Management Team, Score@CET — research development club
- Served as School Captain (2022–2023)

Technical Interests:

- Systems Programming
- Generative AI
- Data Engineering
- Distributed Systems

Languages: English (Fluent) | Malayalam (Fluent) | Hindi (Working Proficiency) | French (Intermediate)